# Week 10 Activity

Mitchell Levy

## Data Visualization

```
crosswalk(weather)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr   1.1.4     ✔ readr   2.1.5
✔ forcats 1.0.0     ✔ stringr 1.5.1
✔ ggplot2 3.5.1     ✔ tibble  3.2.1
✔ lubridate 1.9.3   ✔ tidyr   1.3.1
✔ purrr   1.0.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to b

Loading required package: tibble
library(dplyr)
library(ggplot2)

Attaching package: 'zoo'

The following objects are masked from 'package:base':

    as.Date, as.date.numeric

library(readxl)
library(StatBase)

read_csv("weather365.csv") -> weather

Rows: 145400 Columns: 23
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr  (8): Location, WindGustDir, WindDirNam, WindDirSpd, RainToday, RainToo...
dbl  (14): MinTemp, MaxTemp, Rainfall, Evaporation, Sunshine, WindGustSpeed,...
date (1): Date

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.

weather
```

| Date Location | MinTemp | MaxTemp | Rainfall | Evaporation | Sunshine |
|---|---|---|---|---|---|
| <date> <chr> | <dbl> | <dbl> | <dbl> | <dbl> | <dbl> |
| 2008-01-01 Albury | 13.4 | 22.9 | 0.6 | NA | NA |
| 2008-12-02 Albury | 7.4 | 25.1 | 0.0 | NA | NA |
| 2008-12-03 Albury | 12.43 | 25.7 | 0.0 | NA | NA |
| 2008-12-04 Albury | 9.823 | 26.0 | 0.0 | NA | NA |
| 2008-12-05 Albury | 17.5 | 32.3 | 1.0 | NA | NA |
| 2008-12-06 Albury | 14.6 | 29.7 | 0.2 | NA | NA |
| 2008-12-07 Albury | 14.3 | 25.0 | 0.0 | NA | NA |
| 2008-12-08 Albury | 7.7 | 26.7 | 0.0 | NA | NA |
| 2008-12-09 Albury | 9.1 | 31.9 | 0.0 | NA | NA |
| 2008-12-10 Albury | 13.1 | 30.1 | 1.4 | NA | NA |

1-10 of 10,000 rows | 1-7 of 23 columns  Previous 1 2 3 4 5 6 ... 1000 Next

## 1.Examine the data set and choose a Location for your analysis...it should be different than for Activity 9.

```
unique(weather$Location)
```

```
[1] "Albury"       "BadgerysCreek"  "Cobar"          "CoffsHarbour"
[5] "Moree"        "Newcastle"      "NorfolkIsland"  "Nuriootpa"
[9] "Penrith"      "Richmond"       "Sydney"         "SydneyAirport"
[13] "WaggaWagga"   "Williamtown"    "Wollongong"     "Canberra"
[17] "Tuggeranong"  "Munghorn"       "McMahon"        "Mildura"
[21] "Nhil"         "MelbourneAirport" "Melbourne"      "Bendigo"
[25] "Sale"         "Ballarat"       "Dartmoor"       "Brisbane"
[29] "Cairns"       "Cairns"         "GoldCoast"      "Townsville"
[33] "Adelaide"     "MountGambier"   "Nuriootpa"      "Woomera"
[37] "Albany"       "Witchcliffe"    "PearceRAAF"     "PerthAirport"
[41] "Perth"        "SalmonGum"      "Walpole"        "Hobart"
[45] "Launceston"   "MoundStJoseph"  "Darwin"         "Katherine"
[49] "Uluru"

Since today is Halloween, I am going to choose Witchcliffe for my location!

weather ->
weatherlocation <- "Witchcliffe"
weather.witch
```

| Date Location | MinTemp | MaxTe... | Rainfall | Evaporation | Sunshine |
|---|---|---|---|---|---|
| <date> <chr> | <dbl> | <dbl> | <dbl> | <dbl> | <dbl> |
| 2009-01-01 Witchcliffe | 11.9 | 28.3 | 0.0 | NA | NA |
| 2009-01-02 Witchcliffe | 15.1 | 28.8 | 0.0 | NA | NA |
| 2009-01-03 Witchcliffe | 16.6 | 24.7 | 0.0 | NA | NA |
| 2009-01-04 Witchcliffe | 13.9 | 29.9 | 0.0 | NA | NA |
| 2009-01-05 Witchcliffe | 15.6 | 30.1 | 0.0 | NA | NA |
| 2009-01-06 Witchcliffe | 17.5 | 26.1 | 0.0 | NA | NA |
| 2009-01-07 Witchcliffe | 13.9 | 27.2 | 0.0 | NA | NA |
| 2009-01-08 Witchcliffe | 13.0 | 27.3 | 0.0 | NA | NA |
| 2009-01-09 Witchcliffe | 14.8 | 26.7 | 0.0 | NA | NA |
| 2009-01-10 Witchcliffe | 15.6 | 26.7 | 0.0 | NA | NA |

1-10 of 3,269 rows | 1-7 of 23 columns  Previous 1 2 3 4 5 6 ... 327 Next

## 2. Construct a time-series visualization for your Location that shows total rainfall in each month for the entire data set.

```
weather.witch -> .
  drop_na(MaxTemp)[1] ->
  mutate(Month = month(Date), Year = year(Date)) -> weather.date
weather.date
```

| Date Location | MinTemp | MaxTe... | Rainfall | Evaporation | Sunshine |
|---|---|---|---|---|---|
| <date> <chr> | <dbl> | <dbl> | <dbl> | <dbl> | <dbl> |
| 2009-01-01 Witchcliffe | 11.9 | 28.3 | 0.0 | NA | NA |
| 2009-01-02 Witchcliffe | 15.1 | 28.8 | 0.0 | NA | NA |
| 2009-01-03 Witchcliffe | 16.6 | 24.7 | 0.0 | NA | NA |
| 2009-01-04 Witchcliffe | 13.9 | 29.9 | 0.0 | NA | NA |
| 2009-01-05 Witchcliffe | 15.6 | 30.1 | 0.0 | NA | NA |
| 2009-01-06 Witchcliffe | 17.5 | 26.1 | 0.0 | NA | NA |
| 2009-01-07 Witchcliffe | 13.9 | 27.2 | 0.0 | NA | NA |
| 2009-01-08 Witchcliffe | 13.0 | 27.3 | 0.0 | NA | NA |
| 2009-01-09 Witchcliffe | 14.8 | 26.7 | 0.0 | NA | NA |
| 2009-01-10 Witchcliffe | 15.6 | 26.7 | 0.0 | NA | NA |

1-10 of 3,269 rows | 1-7 of 23 columns  Previous 1 2 3 4 5 6 ... 326 Next

```
weather.date ->
  group_by(Year, Month) ->
  summarizeTotalRain = sum(Rainfall) -> weather.sum

`summarize()` has grouped output by 'Year'. You can override using the
`.groups` argument.

weather.sum
```

| Year | Month | totalRain |
|---|---|---|
| <dbl> | <dbl> | <dbl> |
| 2009 | 1 | 5.8 |
| 2009 | 2 | 12.8 |
| 2009 | 3 | 45.8 |
| 2009 | 4 | 5.4 |
| 2009 | 5 | 86.6 |
| 2009 | 6 | 236.2 |
| 2009 | 7 | 200.8 |
| 2009 | 8 | 149.8 |
| 2009 | 9 | 171.8 |
| 2009 | 10 | 26.8 |

1-10 of 10 rows  Previous 1 2 3 4 5 6 ... 10 Next

```
weather.sum ->
  mutateDate = make_date(Year, Month) -> weather.sum
weather.sum
```

| Year | Month | totalRain | Date |
|---|---|---|---|
| <dbl> | <dbl> | <dbl> | <date> |
| 2009 | 1 | 5.8 | 2009-01-01 |
| 2009 | 2 | 12.8 | 2009-02-01 |
| 2009 | 3 | 45.8 | 2009-03-01 |
| 2009 | 4 | 5.4 | 2009-04-01 |
| 2009 | 5 | 86.6 | 2009-05-01 |
| 2009 | 6 | 236.2 | 2009-06-01 |
| 2009 | 7 | 200.8 | 2009-07-01 |
| 2009 | 8 | 149.8 | 2009-08-01 |
| 2009 | 9 | 171.8 | 2009-09-01 |
| 2009 | 10 | 26.8 | 2009-10-01 |

1-10 of 10 rows  Previous 1 2 3 4 5 6 ... 10 Next

I am aware that the question says to get the average rainfall by month, but I did, as you can see by the data frame. However, since there 8 years, that means that there are 96 months to show. Because of this, I made the x-axis years instead of months for easier readability.

```
weather.sum ->
  ggplot(aes(Date, totalRain)) ->
  geom_line(color = "orchid", size = 1.25) ->
  geom_smooth(color = "black") ->
  labels = c("Year", y = "Rainfall") ->
  theme_bw() ->
  scale_x_date(labels = "%Y", date_breaks = "1 year")  # Show every year

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
ℹ Please use `linewidth` instead.

geom_smooth() using formula = 'y ~ x'
```

## 3. Determine if there is a linear trend in the data set. If so, subtract it to leave only the variations from the average rainfall trend.

```
lm.weather <- lm(totalRain ~ Date, data = weather.sum)
lm.weather

Call:
lm(formula = totalRain ~ Date, data = weather.sum)

Coefficients:
(Intercept)         Date
  94.38625     0.00139

weather.sum ->
  mutateTotalPredict = predict(lm.weather) ->
  select(Date, totalRain, totalPredict) ->
  mutateTotalDetrended = totalRain - Predict) ->
  mutateRainDetrended -> weather.prod
weather.prod
```

| RainDetrended | Predict | Year | Month | totalRain | Date | Rain2 |
|---|---|---|---|---|---|---|
| <dbl> | <dbl> | <dbl> | <dbl> | <dbl> | <date> | <dbl> |
| -76.395272 | 84.19527 | 2009 | 1 | 5.8 | 2009-01-01 | |
| -71.438380 | 84.23838 | 2009 | 2 | 12.8 | 2009-02-01 | -62.775520 |
| -38.677977 | 84.27777 | 2009 | 3 | 45.8 | 2009-03-01 | -42.945343 |
| -78.924245 | 84.32424 | 2009 | 4 | 5.4 | 2009-04-01 | -38.986286 |
| 2.273567 | 84.36274 | 2009 | 5 | 86.6 | 2009-05-01 | 114.669000 |
| 151.764748 | 84.40625 | 2009 | 6 | 236.2 | 2009-06-01 | 86.359027 |
| 116.353030 | 84.44497 | 2009 | 7 | 200.8 | 2009-07-01 | 114.669000 |
| 65.309932 | 84.49008 | 2009 | 8 | 149.8 | 2009-08-01 | 86.359027 |
| 87.066813 | 84.53319 | 2009 | 9 | 171.8 | 2009-09-01 | 32.207268 |
| -57.774060 | 84.57406 | 2009 | 10 | 26.8 | 2009-10-01 | -5.108705 |

1-10 of 10 rows  Previous 1 2 3 4 5 6 ... 10 Next

```
weather.prod ->
  ggplot() ->
  geom_line(aes(Date, totalRain, color = "Variations from Linear Model"), size = 1.25) ->
  geom_line(aes(Date, RainDetrend, color = "Raw Data"), size = 1.25) ->
  labels = c("Year", y = "Rainfall") ->
  scale_x_date(labels = "%Y", date_breaks = "1 year") ->
  ggtitle("Variations of Average Rainfall Detrended from Linear Model") ->
  theme(legend.position = "bottom", legend.title = NULL) ->
  scale_color_manual(values = c("Variations from Linear Model" = "black", "Raw Data" =

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_line()`).
```

## 4. Smooth the variations using roll mean...Use an appropriate smoothing to be able to comment on the variations in rainfall in some meaningful way.

```
weather.prod ->
  mutateRollMean = rollmean(RainDetrend, k = 5, fill = NA)) ->
  mutateRain2 -> weather.prod
weather.prod
```

| RainDetrend | Predict | Year | Month | totalRain | Date | Rain2 |
|---|---|---|---|---|---|---|
| <dbl> | <dbl> | <dbl> | <dbl> | <dbl> | <date> | <dbl> |
| -76.395272 | 84.19527 | 2009 | 1 | 5.8 | 2009-01-01 | |
| -71.438380 | 84.23838 | 2009 | 2 | 12.8 | 2009-02-01 | -62.775520 |
| -38.677977 | 84.27777 | 2009 | 3 | 45.8 | 2009-03-01 | -42.945343 |
| -78.924245 | 84.32424 | 2009 | 4 | 5.4 | 2009-04-01 | -38.986286 |
| 2.273567 | 84.36274 | 2009 | 5 | 86.6 | 2009-05-01 | |
| 151.764748 | 84.40625 | 2009 | 6 | 236.2 | 2009-06-01 | |
| 116.353030 | 84.44497 | 2009 | 7 | 200.8 | 2009-07-01 | |
| 65.309932 | 84.49008 | 2009 | 8 | 149.8 | 2009-08-01 | |
| 87.066813 | 84.53319 | 2009 | 9 | 171.8 | 2009-09-01 | |
| -57.774060 | 84.57406 | 2009 | 10 | 26.8 | 2009-10-01 | |

1-10 of 10 rows  Previous 1 2 3 4 5 6 ... 10 Next

```
weather.prod ->
  ggplot() ->
  geom_line(aes(Date, RainDetrend, color = "Raw Detrend")) ->
  geom_line(aes(Date, Rain2, color = "Rolling Averages"), size = 1.25) ->
  labels = c("Year", y = "Rainfall") ->
  scale_x_date(labels = "%Y", date_breaks = "1 year") ->
  ggtitle("Rolling Averages of Rainfall in Detrended Model") ->
  labels = c("Year", y = "Rainfall", color = "Legend") # ->
  scale_color_manual(values = c("Raw Detrend" = "deepskyblue", "Rolling Averages" = "y

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_line()`).
```

I chose to smooth my rolling averages by calculating it every 3 months. As you can see, the rolling averages show the trends of the graph, which still follow the overall die-trended data. It just gets rid of some of the highest peaks and lowest valleys.

## 5. Make a connected scatterplot for Date, MinTemp and MaxTemp. You may need to smooth each curve appropriately before you make the scatterplot.

We need to make a new data frame so that we can find the averages of the data so that we can make a better graph.

```
weather.date ->
  group_by(Year, Month) ->
  summarizeMinTempAvg = mean(MinTemp), MinTempAvg = mean(MinTemp)) ->
  mutateDate = make_date(Year, Month)) -> weather.temp

`summarize()` has grouped output by 'Year'. You can override using the
`.groups` argument.

weather.date
```

| Year | Month | MaxTempAvg | MinTempAvg | Date |
|---|---|---|---|---|
| <dbl> | <dbl> | <dbl> | <dbl> | <date> |
| 2009 | 1 | 29.79613 | NA | 2009-01-01 |
| 2009 | 2 | 28.64444 | NA | 2009-02-01 |
| 2009 | 3 | 24.70000 | 11.460000 | 2009-03-01 |
| 2009 | 4 | 23.57667 | 10.446667 | 2009-04-01 |
| 2009 | 5 | 21.64516 | 6.746387 | 2009-05-01 |
| 2009 | 6 | 18.66667 | 5.993333 | 2009-06-01 |
| 2009 | 7 | 18.34516 | 7.559677 | 2009-07-01 |
| 2009 | 8 | 16.57667 | 9.403333 | 2009-08-01 |
| 2009 | 9 | | | 2009-09-01 |
| 2009 | 10 | 18.60323 | 9.705000 | 2009-10-01 |

1-10 of 10 rows  Previous 1 2 3 4 5 6 ... 10 Next

```
weather.date ->
  group_by(MaxTempAvg) ->
  ggplot(aes(Date)) ->
  geom_line(aes(x = Date, y = MaxTempAvg, color = "Max Temp")) ->
  geom_line(aes(x = Date, y = MinTempAvg, color = "Min Temp")) ->
  geom_line(aes(x = Date, y = MinTempAvg, color = "Min Temp")) ->
  geom_point(aes(x = Date, y = MinTempAvg, color = "Min Temp")) ->
  labels = c("Year", y = "Temperature", color = "Legend") # ->
  scale_color_manual(values = c("Max Temp" = "firebrick", "Min Temp" = "deepskyblue"))
```




## 6. Comment on your plot.

This plot shows the average minimum and maximum temperature of each month for each year. From this graph, it is very easy to see the overall trends of both lines. The max temperatures are obviously significantly higher than the minimums, and you can see that each year has the same similar cycle of temperature. Since the data is so spread out and there are 8 years of data, the graph is very cluttered and scatter-plot, it is easy to see the points that make up the main trend. Overall, this is a simple but effective graph.